

BUSAN 300
Project Report

Team 31

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1. Project Summary

This project examined the relationship between GDP per capita and suicide rates across countries in 2020 using a combined dataset containing GDP and suicide-rate information. The purpose of the project was to determine whether economic differences between countries were associated with differences in suicide rates and suicide-rate gender gaps.

The first question asked: Do countries with higher GDP per capita tend to have higher or lower suicide rates in 2020? The analysis found a weak positive relationship, meaning countries with higher GDP per capita tended to have slightly higher suicide rates. However, the relationship was weak, suggesting that GDP per capita alone does not strongly explain differences in suicide rates across countries.

The second question asked: Do high-GDP-per-capita countries have a larger or smaller gender gap in suicide rates than low-GDP-per-capita countries in 2020? The results showed that high-GDP-per-capita countries generally had larger gender gaps in suicide rates than low-GDP-per-capita countries.

The third question asked: Which countries deviate most from the expected relationship between GDP per capita and suicide rates in 2020? Several countries, including Lesotho, Guyana, Suriname, South Korea, Swaziland, Lithuania, Ukraine, Uruguay, South Africa and Russia, had suicide rates that differed considerably from what would be expected based on GDP per capita alone.

Overall, the findings suggest that GDP per capita has limited ability to explain differences in suicide rates and gender gaps across countries. While some relationships were identified, the results indicate that other factors, such as healthcare access, social conditions, cultural differences and demographic patterns, may also influence suicide rates. Therefore, GDP per capita should be considered only one of several factors when examining suicide rates across countries.

Research Question	Key Findings
Do countries with higher GDP per capita tend to have higher or lower suicide rates in 2020?	Weak positive relationship
Do high-GDP-per-capita countries have a larger or smaller gender gap in suicide rates than low-GDP-per-capita countries in 2020?	High-GDP-per-capita countries had larger gender gaps
Which countries deviate most from the expected relationship between GDP per capita and suicide rates in 2020?	Lesotho, Guyana, Suriname, South Korea, Swaziland, Lithuania, Ukraine, Uruguay, South Africa and Russia

2. Wrangling Details

2.1 Dataset selection justification

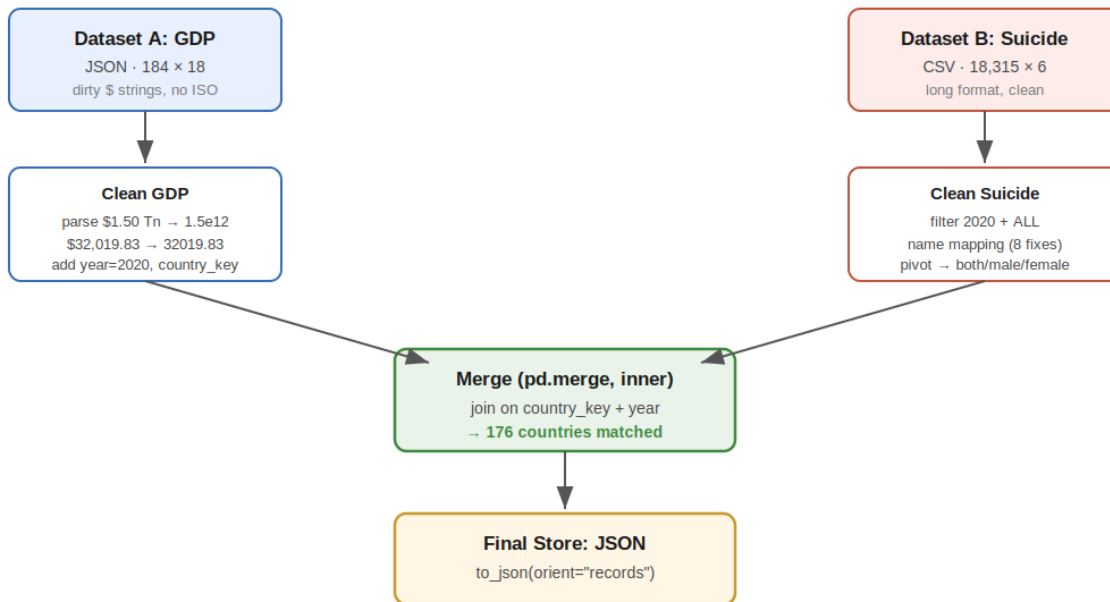
These two datasets were selected because together they enable questions that neither could answer alone, directly addressing the project theme of a sustainable and equitable society. Suicide rate is a sensitive indicator of population mental-health and social wellbeing, while GDP per capita is a standard measure of national economic development; combining them allows us to examine whether economic prosperity is associated with mental-health outcomes and gender inequality a question of clear social relevance.

The specific datasets were preferred over alternatives for three reasons. First, format compatibility: one is JSON and the other CSV, satisfying the requirement for two different formats, and they share a clean common key (country name) that makes a meaningful merge possible. Second, coverage and granularity: the suicide dataset spans 185 countries with breakdowns by sex and age group, which supports a richer gender gap analysis than a single aggregate figure would, while the GDP dataset provides comparable country-level economic indicators for the same set of nations. Third, source reliability: both originate from Kaggle datasets compiled from recognised international statistics, and the GDP dataset was provided through the course Canvas page, satisfying the requirement that at least one dataset come from the provided list.

We also considered using only a single economic indicator (e.g. total GDP), but per-capita GDP was chosen because it normalises for population size and is more comparable across countries of very different sizes.

Before we start to think about some questions, we need to clean each dataset first and ensure that the two datasets merge well. Also, ensure that the two datasets are consistent and formatted similarly, such as the country's names. Avoid duplication, which can lead to low accuracy in our answer. All cleaning and combining were performed in Python (pandas) on Google Colab. No file converters or automation tools were used. Every transformation is written explicitly in code and is therefore fully replicable. The combined dataset was exported to a single JSON file as the final data store.

Data Wrangling Pipeline



All steps in Python/pandas on Google Colab — no converters or automation tools.

2.2 The Datasets

2.2.1 Dataset A — Global GDP (JSON)

Origin. A JSON file of 2020 economic and demographic indicators for countries worldwide, sourced from Kaggle (“World GDP” dataset). Direct link:

<https://www.kaggle.com/datasets/darknez/gdp-among-world>

General characteristics:

- **Format:** JSON. Read with `pd.read_json()` into a 184×18 table.
- **Records:** 184 countries.
- **Column:** 18, including Country, GDP Per Capita, GDP (IMF), GDP (UN), Region, Subregion, Population Rank, Area, Capital City.
- **Join key:** Country (country name). The file contains no ISO country code.

Initial audit — observations and problems found:

- **Non-uniform structure (missing values).** The GDP dataset had several data quality issues that needed to be addressed before combining it with the suicide dataset. First, the dataset contained missing values across some fields. For example, Anthem was missing in 158 out of 184 records, and Government was missing in 106 records. However, these fields were not required for our analysis. Some relevant fields, such as Population Rank, Growth Rate, Land Area, and World Percentage, each had 2 missing values. No duplicate rows were found.

- **Dirty numeric values stored as strings.**

Several numeric values were stored as text strings rather than numeric data. For example, GDP (IMF) appeared in formats such as "\$1.50 Tn", "\$26.35 Bn", and "\$40.71 Mn", while GDP Per Capita contained dollar signs and thousands of separators, such as "\$32,019.83". These values had to be cleaned and converted into plain numeric values before calculation or comparison.

- **Single-year snapshot.** GDP represents 2020 only (see limitation in §3).

The GDP dataset was a single-year snapshot for 2020. As a result, the suicide dataset also had to be filtered to 2020 so that both datasets represented the same year.

```
# Read the two raw datasets
gdp_raw = pd.read_json("GDPfinal.json")
suicide_raw = pd.read_csv("suicide.csv")

# Check basic structure
print("GDP shape:", gdp_raw.shape)
print("Suicide shape:", suicide_raw.shape)

print("\nGDP columns:")
print(gdp_raw.columns)

print("\nSuicide columns:")
print(suicide_raw.columns)

display(gdp_raw.head())
display(suicide_raw.head())
```

```
... GDP shape: (184, 18)
Suicide shape: (18315, 6)
```

```
GDP columns:
Index(['Country', 'Population Rank', 'Growth Rate', 'World Percentage',
      'Density', 'Land Area', '2020 Population Rank', '2020 World Percentage',
      '2020 Growth Rate', 'Area', 'Capital City', 'Region', 'Subregion',
      'Anthem', 'Government', 'GDP (IMF)', 'GDP (UN)', 'GDP Per Capita'],
      dtype='object')
```

```
Suicide columns:
Index(['iso_code', 'country', 'year', 'sex', 'age_group', 'suicide_rate'], dtype='object')
```

	Country	Population Rank	Growth Rate	World Percentage	Density	Land Area	2020 Population Rank	2020 World Percentage	2020 Growth Rate	Area	Capital City	Region	Subregion	Anthem	Government	GDP (IMF)	GDP (UN)	GDP Per Capita
0	Spain	30.0	0.04%	0.60%	92.40	498,800	30	0.60%	0.04%	505,992	Madrid	Europe	Southern Europe	Spain	Spain	\$1.50 Tn	\$1.24 Tn	\$32,019.83
1	Cyprus	158.0	0.73%	0.02%	130.51	9,240	158	0.02%	0.73%	9,251	Nicosia	Europe	Eastern Europe	NaN	NaN	\$26.35 Bn	\$20.05 Bn	\$21,822.84
2	Bahrain	152.0	3.68%	0.02%	2224.28	760	152	0.02%	3.68%	765	Manama	Asia	Western Asia	NaN	NaN	\$40.71 Bn	\$32.18 Bn	\$23,923.13
3	Bhutan	165.0	1.12%	0.01%	20.10	38,117	165	0.01%	1.12%	38,394	Thimphu	Asia	Southern Asia	NaN	NaN	\$2.94 Bn	\$2.21 Bn	\$3,808.93
4	Singapore	114.0	0.79%	0.08%	8239.92	700	114	0.08%	0.79%	710	Singapore	Asia	South-Eastern Asia	NaN	NaN	\$391.88 Bn	\$296.95 Bn	\$66,983.26

2.2.2 Dataset B — Global Suicide Rates 2000–2021 (CSV)

Origin. A CSV file of suicide rates by country, year, sex and age group, sourced from Kaggle (“Global Suicide Rates by Country 2000–2021”). Direct link:

<https://www.kaggle.com/datasets/lucalullo/global-suicide-rates-by-country-2000-2021>

General characteristics:

- **Format:** CSV. Read with `pd.read_csv()` into an $18,315 \times 6$ table.
- **Records:** 18,315 rows.
- **Columns:** 6 — `iso_code`, `country`, `year`, `sex`, `age_group`, `suicide_rate`.
- **Coverage:** 185 countries $\times$ years 2000–2021 $\times$ 3 sexes (both, male, female) $\times$ 12 age groups.
- **Join key:** `country` (country name); an `iso_code` is also present.

Initial audit — observations and problems found:

- **The data is clean:** The suicide dataset was relatively clean compared with the GDP dataset. The initial audit found 0 missing values and 0 duplicate rows, and the year range was confirmed as 2000–2021.
- **Long format.** Each country is split across many rows (by year/sex/age group). We filter to `age_group = "ALL"` and `year = 2020`, then reshape (see §2.2). The dataset was stored in a long format. This means that each country appeared across multiple rows because the data was separated by year, sex, and `age_group`. Since the GDP dataset only represents 2020, the suicide dataset had to be filtered to `year = 2020` before merging. It was also filtered to `age_group = "ALL"` so that the final dataset represented overall suicide rates rather than separate age categories.

```

# Read the two raw datasets
gdp_raw = pd.read_json("GDPfinal.json")
suicide_raw = pd.read_csv("suicide.csv")

# Check basic structure
print("GDP shape:", gdp_raw.shape)
print("Suicide shape:", suicide_raw.shape)

print("\nGDP columns:")
print(gdp_raw.columns)

print("\nSuicide columns:")
print(suicide_raw.columns)

display(gdp_raw.head())
display(suicide_raw.head())

```

	iso_code	country	year	sex	age_group	suicide_rate
0	AFG	Afghanistan	2000	male	ALL	5.793457
1	AFG	Afghanistan	2000	both	ALL	4.356148
2	AFG	Afghanistan	2000	female	ALL	2.910393
3	AFG	Afghanistan	2001	male	ALL	5.831328
4	AFG	Afghanistan	2001	both	ALL	4.383031

How the two datasets relate (and merge feasibility))

The datasets share country names as their common attribute, which is the basis for combining them. An audit of the names found that a direct match on lower-cased names links most countries, but about 8 of the same countries are spelled differently between the two files and would be lost without intervention (e.g. Czechia vs Czech Republic, Cabo Verde vs Cape Verde, Côte d'Ivoire vs Ivory Coast). A small number of countries (e.g. Cuba, North Korea, Syria) are genuinely absent from the GDP file and are dropped by the inner join. After name alignment, 176 countries match between the two datasets.

2. 3 Wrangling Process — Cleaning and Combining

2.3.1 Transformations on Dataset A (GDP)

1. **Parsed dirty monetary strings into numbers** with a custom function `money_to_number()`. It removes the \$ sign and thousands-separator commas, then uses a regular expression to split the value into a number and a unit, multiplying by the unit's scale (Tn = 10^{12} , Bn = 10^9 , Mn = 10^6). Examples: "\$1.50 Tn" → $1.5e12$; "\$32,019.83" → 32019.83. Applied to GDP (IMF), GDP (UN), and GDP Per Capita.
2. **Added year = 2020** so the single GDP snapshot can be joined on year.
3. **Built a join key** `country_key` = country name, stripped of whitespace and lower-cased.
4. **Kept only the needed fields** (Country, `country_key`, year, Region, Subregion, and the three numeric GDP columns). Result: A clean 184×8 table.

```
# Make a clean copy of the GDP dataset
gdp_clean = gdp_raw.copy()

# The GDP dataset is treated as 2020 country-level data
gdp_clean["year"] = 2020

# Create a simple country key for matching
gdp_clean["country_key"] = (
    gdp_clean["Country"]
    .astype(str)
    .str.strip()
    .str.lower()
)

# Convert GDP fields from text to numbers
gdp_clean["gdp_imf_usd"] = gdp_clean["GDP (IMF)"].apply(money_to_number)
gdp_clean["gdp_un_usd"] = gdp_clean["GDP (UN)"].apply(money_to_number)
gdp_clean["gdp_per_capita_usd"] = gdp_clean["GDP Per Capita"].apply(money_to_number)

# Keep useful fields only
gdp_clean = gdp_clean[
    [
        "Country",
        "country_key",
        "year",
        "Region",
        "Subregion",
        "gdp_imf_usd",
        "gdp_un_usd",
        "gdp_per_capita_usd"
    ]
]

display(gdp_clean.head())

print("GDP clean shape:", gdp_clean.shape)
print("\nMissing values after cleaning:")
print(gdp_clean.isnull().sum())
```

Country	country_key	year	Region	Subregion	gdp_imf_usd	gdp_un_usd	gdp_per_capita_usd
Spain	spain	2020	Europe	Southern Europe	1.500000e+12	1.240000e+12	32019.83
Cyprus	cyprus	2020	Europe	Eastern Europe	2.635000e+10	2.005000e+10	21822.84
Bahrain	bahrain	2020	Asia	Western Asia	4.071000e+10	3.218000e+10	23923.13
Bhutan	bhutan	2020	Asia	Southern Asia	2.940000e+09	2.210000e+09	3808.93
Singapore	singapore	2020	Asia	South-Eastern Asia	3.918800e+11	2.969500e+11	66983.26

GDP clean shape: (184, 8)

Missing values after cleaning:

Country	0
country_key	0
year	0
Region	0
Subregion	0
gdp_imf_usd	2
gdp_un_usd	2
gdp_per_capita_usd	0

dtype: int64

2.3.2 Transformations on Dataset B (suicide)

- Filtered** to year == 2020 and age_group == "ALL".
- Aligned country names** using a manual mapping table (applied before building the join key), so spelling differences do not break the merge:

Suicide CSV name	GDP JSON name
Cabo Verde	Cape Verde
Czechia	Czech Republic
Côte d'Ivoire	Ivory Coast
Democratic Republic of the Congo	DR Congo
North Macedonia	Macedonia
São Tomé and Príncipe	Sao Tome and Principe
Eswatini	Swaziland
Türkiye	Turkey

Country names were manually aligned before creating the country_key. This was necessary because some countries had different names in the two datasets. For example, the suicide dataset used Türkiye, while the GDP dataset used Turkey. Without this mapping, these countries would not match during the merge.

- Built the same country_key** (stripped, lower-cased).
- Ensured suicide_rate is numeric** with pd.to_numeric().
- Reshaped from long to wide** with pivot_table(), turning the sex rows into three columns: suicide_rate_both, suicide_rate_female, suicide_rate_male. A gender_gap column (male – female) was also computed. Result: one row per country (185 rows), 0 missing values.

Explanation: Built the same standardised country_key in the suicide dataset by removing extra spaces and converting country names to lowercase. The suicide_rate column was converted to numeric format using pd.to_numeric() so it could be used for calculation. The dataset was then reshaped from a long format to a wide format using pivot_table(). This turned the sex values into separate columns: suicide_rate_both, suicide_rate_female, and suicide_rate_male. A new gender_gap column was also calculated as male suicide rate minus female suicide rate. After reshaping, the dataset contained one row per country for 2020, with 185 rows and 0 missing values.

```

62] # Put male, female and both suicide rates into separate columns
    suicide_pivot = suicide_2020.pivot_table(
        index=["iso_code", "country", "year", "country_key"],
        columns="sex",
        values="suicide_rate",
        aggfunc="first"
    ).reset_index()

    # Rename columns clearly
    suicide_pivot = suicide_pivot.rename(columns={
        "both": "suicide_rate_both",
        "female": "suicide_rate_female",
        "male": "suicide_rate_male"
    })

    # Calculate gender gap
    suicide_pivot["gender_gap"] = (
        suicide_pivot["suicide_rate_male"] - suicide_pivot["suicide_rate_female"]
    )

    display(suicide_pivot.head())

    print("Suicide pivot shape:", suicide_pivot.shape)
    print("\nMissing values:")
    print(suicide_pivot.isnull().sum())

```

	sex	iso_code	country	year	country_key	suicide_rate_both	suicide_rate_female	suicide_rate_male	gender_gap
0		AFG	Afghanistan	2020	afghanistan	3.493914	2.394483	4.572058	2.177576
1		AGO	Angola	2020	angola	7.409206	3.415952	11.492239	8.076288
2		ALB	Albania	2020	albania	3.355071	2.291795	4.440141	2.148346
3		ARE	United Arab Emirates	2020	united arab emirates	2.377881	0.400705	3.471652	3.070946
4		ARG	Argentina	2020	argentina	8.121802	3.115624	13.206801	10.091178

Suicide pivot shape: (185, 8)

Missing values:

```

sex
iso_code      0
country       0
year          0
country_key   0
suicide_rate_both  0
suicide_rate_female  0
suicide_rate_male  0
gender_gap    0
dtype: int64

```

2.3.3 Combining the datasets

- The two cleaned tables were merged with pd.merge() on ["country_key", "year"] using an inner join.
- After the name alignment in §2.2, 176 countries matched. Countries with no GDP counterpart (Cuba, North Korea, Syria, etc.) are dropped by the inner join — expected and documented.

Explanation: The two cleaned tables were merged using `pd.merge()` on `country_key` and `year` with an inner join. These two fields were used together to ensure that each GDP record was matched with the suicide rate record for the same country and year. Because an inner join only keeps records that exist in both datasets, countries that appeared in the suicide dataset but had no matching GDP record, such as Cuba, North Korea and Syria, were excluded from the final combined dataset. After the country-name alignment, 176 countries were successfully matched.

```
# Test merge using country name and year
combined_data = pd.merge(
    gdp_clean,
    suicide_pivot,
    on=["country_key", "year"],
    how="inner"
)

print("GDP rows:", len(gdp_clean))
print("Suicide rows:", len(suicide_pivot))
print("Matched rows before mapping:", len(combined_data))

display(combined_data.head())
```

... : 184
rows: 185
rows before mapping: 176

try	country_key	year	Region	Subregion	gdp_imf_usd	gdp_un_usd	gdp_per_capita_usd	iso_code	country	suicide_rate_both	suicide_rate_female	suicide_rate_male	gender_gap
pain	spain	2020	Europe	Southern Europe	1.500000e+12	1.240000e+12	32019.83	ESP	Spain	8.624267	4.518217	12.878482	8.360265
prus	cyprus	2020	Europe	Eastern Europe	2.635000e+10	2.005000e+10	21822.84	CYP	Cyprus	3.864847	2.068085	5.631349	3.563264
rain	bahrain	2020	Asia	Western Asia	4.071000e+10	3.218000e+10	23923.13	BHR	Bahrain	5.478217	1.179637	8.078285	6.898648
utan	bhutan	2020	Asia	Southern Asia	2.940000e+09	2.210000e+09	3808.93	BTN	Bhutan	5.076827	3.408273	6.517589	3.109316
ore	singapore	2020	Asia	South-Eastern Asia	3.918800e+11	2.969500e+11	66983.26	SGP	Singapore	8.938073	5.419005	12.216388	6.797383

2.3.4 Storing the combined dataset (final storage format)

- The combined DataFrame was exported to a single JSON file (`combined_suicide_gdp.json`) with `DataFrame.to_json(orient="records", indent=2, force_ascii=False)`.
- This is the project's final data store: each JSON object is one country and now contains, in one place, information that originally lived in two separate files of two different formats (CSV + JSON).
- Note: this export is a storage step performed in code on already-cleaned data; it is not a file-conversion tool

```

# === Export the combined dataset as JSON (final storage format) ===

# Perform the final merge on the name-aligned data (should give 176 countries)
combined = pd.merge(
    gdp_clean,
    suicide_pivot,
    on=["country_key", "year"],
    how="inner"
)
print("Final combined rows:", len(combined)) # should be 176

# Export to a single JSON file
combined.to_json(
    "combined_suicide_gdp.json",
    orient="records",
    indent=2,
    force_ascii=False
)
print("Exported combined_suicide_gdp.json")

# Preview the first two records to confirm structure (use this for the screenshot)
import json
with open("combined_suicide_gdp.json") as f:
    data = json.load(f)
print("Total records:", len(data))
print(json.dumps(data[:2], indent=2, ensure_ascii=False))

```

Explanation: The final combined DataFrame was exported to a single JSON file named `combined_suicide_gdp.json` using `DataFrame.to_json(orient="records", indent=2, force_ascii=False)`. The records orientation was used so that each JSON object represents one country in 2020, containing both GDP and suicide rate information in the same record.

The parameters `indent=2` and `force_ascii=False` were used to make the file readable and to preserve country names with special characters. This JSON file is the project's final data store because it contains cleaned and combined data from the original CSV and JSON sources.

This export was completed through explicit Python code after the data had already been cleaned and merged. Therefore, it was used as a storage step rather than as a file conversion or automation tool.

```

... Final combined rows: 176
Exported combined_suicide_gdp.json
Total records: 176
[
  {
    "Country": "Spain",
    "country_key": "spain",
    "year": 2020,
    "Region": "Europe",
    "Subregion": "Southern Europe",
    "gdp_imf_usd": 150000000000.0,
    "gdp_un_usd": 124000000000.0,
    "gdp_per_capita_usd": 32019.83,
    "iso_code": "ESP",
    "country": "Spain",
    "suicide_rate_both": 8.624266997,
    "suicide_rate_female": 4.518217055,
    "suicide_rate_male": 12.87848228,
    "gender_gap": 8.360265225
  },
  {
    "Country": "Cyprus",
    "country_key": "cyprus",
    "year": 2020,
    "Region": "Europe",
    "Subregion": "Eastern Europe",
    "gdp_imf_usd": 26350000000.0,
    "gdp_un_usd": 20050000000.0,
    "gdp_per_capita_usd": 21822.84,
    "iso_code": "CYP",
    "country": "Cyprus",
    "suicide_rate_both": 3.86484692,
    "suicide_rate_female": 2.068084833,
    "suicide_rate_male": 5.63134908,
    "gender_gap": 3.563264247
  }
]

```

2.4 Limitation / Key Assumption

GDP is a single-year (2020) snapshot, and the analysis is therefore restricted to the 2020 cross-section of the suicide data. Relationships found describe 2020 only and should not be read as trends over time or as causal effects.

2.5 Link to the Final Data Store

<https://drive.google.com/file/d/1KQKjOFwxounasgSSoq0ttFTy-AkffXDJ/view?usp=sharing>

3. Questions and Answers

Question 01: Do countries with higher GDP per capita tend to have higher or lower suicide rates in 2020?

Answer:

Countries with higher GDP per capita tended to have slightly higher suicide rates in 2020. However, the relationship was weak. Therefore, GDP per capita alone does not strongly explain differences in suicide rates across countries.

Data used:

This question was answered using the final combined data store, combined_suicide_gdp.json. The file contains one record per matched country in 2020. For this question, we used two fields from the final JSON file: gdp_per_capita_usd and suicide_rate_both.

Steps and tools used:

The analysis was performed in Google Colab using Python, pandas and Matplotlib.

First, the final JSON data store was loaded into a pandas DataFrame using `pd.read_json("combined_suicide_gdp.json")`. We then selected `gdp_per_capita_usd` as the economic variable and `suicide_rate_both` as the overall suicide rate variable.

To measure the relationship, we calculated both Pearson and Spearman correlations. Pearson correlation was used to measure the linear relationship between GDP per capita and suicide rate. Spearman correlation was also calculated because GDP per capita is highly uneven across countries and a rank-based measure can provide an additional check.

The key operations executed were:

```
df = pd.read_json("combined_suicide_gdp.json")
```

```
pearson_corr = df["gdp_per_capita_usd"].corr(  
    df["suicide_rate_both"],  
    method="pearson"  
)
```

```
spearman_corr = df["gdp_per_capita_usd"].corr(  
    df["suicide_rate_both"],  
    method="spearman"  
)
```

Two scatter plots were created to visually inspect the relationship. Figure 1 shows the relationship using the original GDP per capita scale, while Figure 2 shows GDP per capita on a log scale. The log-scale version was included because GDP per capita values vary greatly across countries and the log scale makes the distribution easier to interpret.

Results:

The Pearson correlation was 0.190, and the Spearman correlation was 0.236. Both values are positive, but small. This means there was a weak positive association between GDP per capita and suicide rate in the 2020 combined dataset.

Figures 1 and 2 show that the data points are widely spread out rather than forming a clear upward or downward pattern. Figure 2 is easier to interpret because GDP per capita is shown on a log scale. Some higher-GDP-per-capita countries had moderate or high suicide rates, while some lower-GDP-per-capita countries also had high suicide rates. This supports the conclusion that the relationship is weak.

Interpretation:

Based on the final combined dataset, countries with higher GDP per capita tended to have slightly higher suicide rates in 2020. However, the weak correlation and dispersed scatter plot show that GDP per capita is not a strong predictor of suicide rates. This suggests that economic wealth alone cannot fully explain national suicide rates. Other factors, such as healthcare access, social conditions, cultural differences, reporting practices and demographic patterns, may also influence suicide rates.

Limitation:

This analysis uses only 2020 cross-sectional data. It describes an association between GDP per capita and suicide rate in one year only. It should not be interpreted as evidence that GDP per capita causes suicide rates to increase or decrease.

Measure	Value	Interpretation
Pearson correlation	0.19	Weak positive linear relationship
Spearman correlation	0.236	Weak positive rank-based relationship

Table 1. Correlation between GDP per capita and overall suicide rate in 2020.

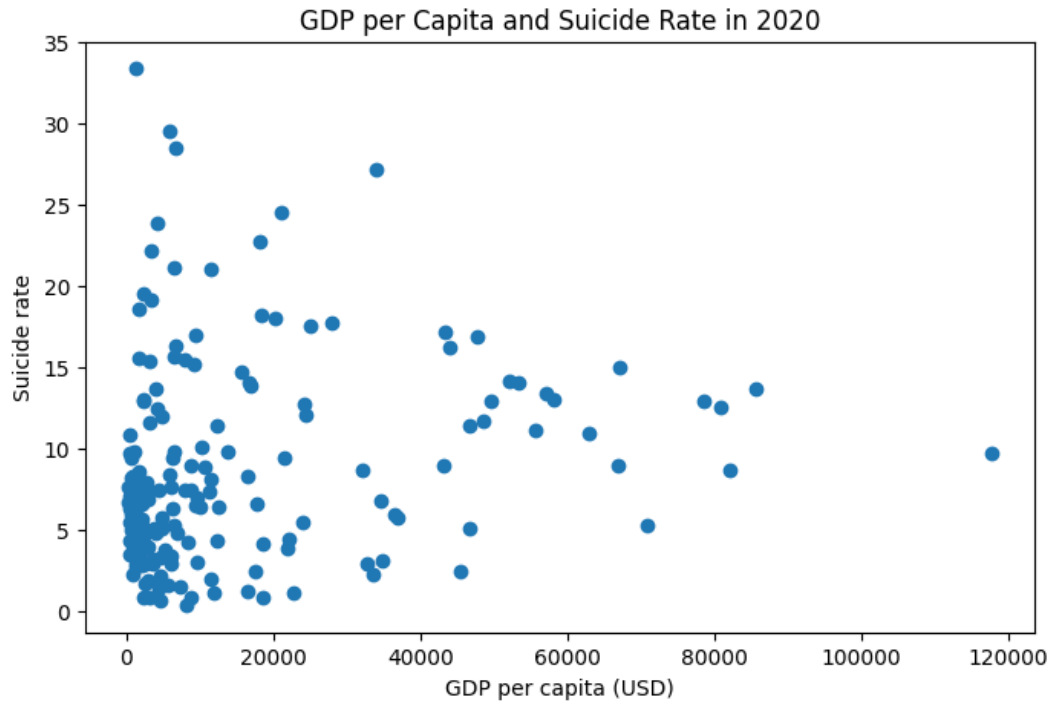


Figure 1. Relationship between GDP per capita and overall suicide rate in 2020.

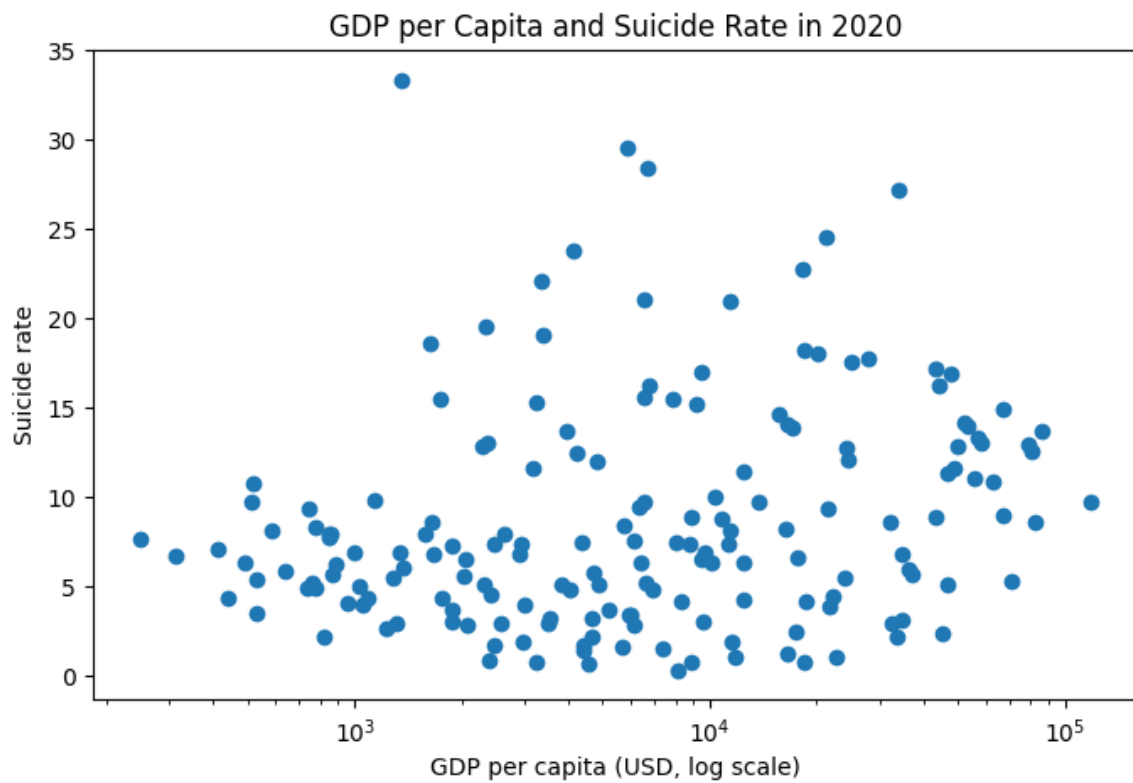


Figure 2. Relationship between GDP per capita and overall suicide rate in 2020(log-scale).

Question 02: Do high-GDP-per-capita countries have a larger or smaller gender gap in suicide rates than low-GDP-per-capita countries in 2020?

Answer:

High-GDP-per-capita countries had a larger average gender gap in suicide rates than low-GDP-per-capita countries in 2020. In the final combined dataset, the average gender gap was 10.451 for the high-GDP-per-capita group and 6.886 for the low-GDP-per-capita group. The difference between the two groups was 3.565 suicide-rate points.

Data used:

This question was answered using the final combined JSON data store, `combined_suicide_gdp.json`. The file contains one record per matched country in 2020. The main fields used were:

- `gdp_per_capita_usd`
- `suicide_rate_male`
- `suicide_rate_female`
- `gender_gap`

The `gender_gap` field was defined as:

`gender_gap = suicide_rate_male - suicide_rate_female`

This means a larger value indicates a larger difference between male and female suicide rates.

Steps and tools used:

The analysis was performed in Google Colab using Python, pandas, NumPy and Matplotlib.

First, the final JSON data store was loaded into a pandas DataFrame:

```
df = pd.read_json("combined_suicide_gdp.json")
```

We then checked that the key columns needed for this question had no missing values. We also recalculated the gender gap to confirm that the existing `gender_gap` column was correct:

```
df["gender_gap_check"] = df["suicide_rate_male"] - df["suicide_rate_female"]  
difference = (df["gender_gap"] - df["gender_gap_check"]).abs().max()
```

The maximum difference was approximately zero, confirming that `gender_gap` was correctly calculated as male suicide rate minus female suicide rate.

Next, we defined high- and low-GDP-per-capita countries using quartiles. Countries in the top 25% of GDP per capita were classified as the high-GDP-per-capita group, and countries in the bottom 25% were classified as the low-GDP-per-capita group. The middle 50% were excluded from this comparison so that the analysis compared clearly different economic groups.

The cut-off values were:

- Low-GDP-per-capita group: GDP per capita at or below 2,052.2775 USD
- High-GDP-per-capita group: GDP per capita at or above 18,416.8675 USD

This produced:

- 44 countries in the low-GDP-per-capita group

- 44 countries in the high-GDP-per-capita group
- 88 countries in the middle group

The key operation was:

```
low_cutoff = df["gdp_per_capita_usd"].quantile(0.25)
high_cutoff = df["gdp_per_capita_usd"].quantile(0.75)
```

```
df["gdp_group"] = np.where(
    df["gdp_per_capita_usd"] <= low_cutoff,
    "Low GDP per capita",
    np.where(
        df["gdp_per_capita_usd"] >= high_cutoff,
        "High GDP per capita",
        "Middle"
    )
)
```

We then grouped the high- and low-GDP countries and calculated the average male suicide rate, average female suicide rate, average overall suicide rate, average gender gap and median gender gap.

Results:

GDP per capita group	Countries	Average GDP per capita	Average suicide rate	Average male suicide rate	Average female suicide rate	Average gender gap	Median gender gap
High GDP per capita	44	44,815.63	10.448	15.699	5.249	10.451	8.928
Low GDP per capita	44	1,046.00	7.24	10.729	3.843	6.886	5.611

Table 2. Comparison of suicide-rate gender gap between high- and low-GDP-per-capita countries in 2020.

The high-GDP-per-capita group had an average gender gap of 10.451, while the low-GDP-per-capita group had an average gender gap of 6.886. The difference was 3.565 suicide-rate points.

Interpretation:

The result shows that high-GDP-per-capita countries had a larger gender gap in suicide rates than low-GDP-per-capita countries in 2020. This means that, in the high-GDP-per-capita group, the difference between male and female suicide rates was larger on average.

The bar chart supports this result by showing a higher average gender gap for the high-GDP-per-capita group. The boxplot also shows that the high-GDP-per-capita group generally had a higher distribution of gender gaps, although there were outliers in both groups.

This finding suggests that higher national income levels do not necessarily correspond to smaller gender differences in suicide rates. However, GDP per capita alone cannot explain why the gender gap is larger. Other social, cultural, healthcare, demographic and reporting factors may also influence the observed gender gap.

Limitation:

This comparison is based only on 2020 cross-sectional data. It shows an observed difference between high- and low-GDP-per-capita groups, but it does not prove that GDP per capita causes a larger or smaller gender gap. The high and low groups were defined using quartiles within this dataset, so the result should be interpreted as a comparison within the final combined dataset rather than a universal rule.

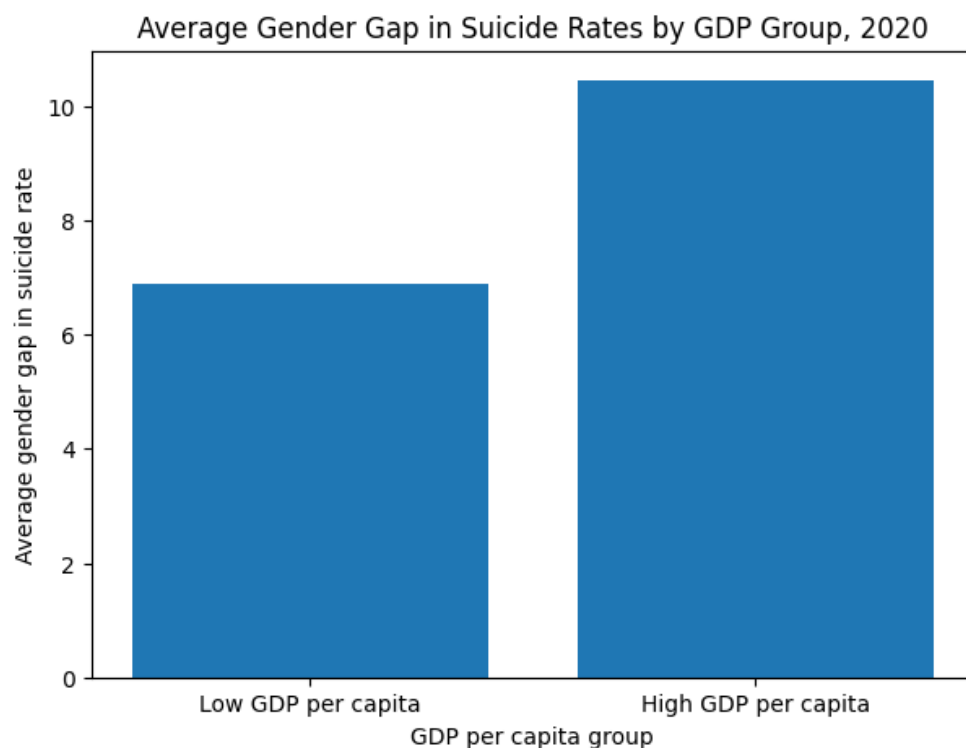


Figure 3. Average gender gap in suicide rates by GDP per capita group, 2020.

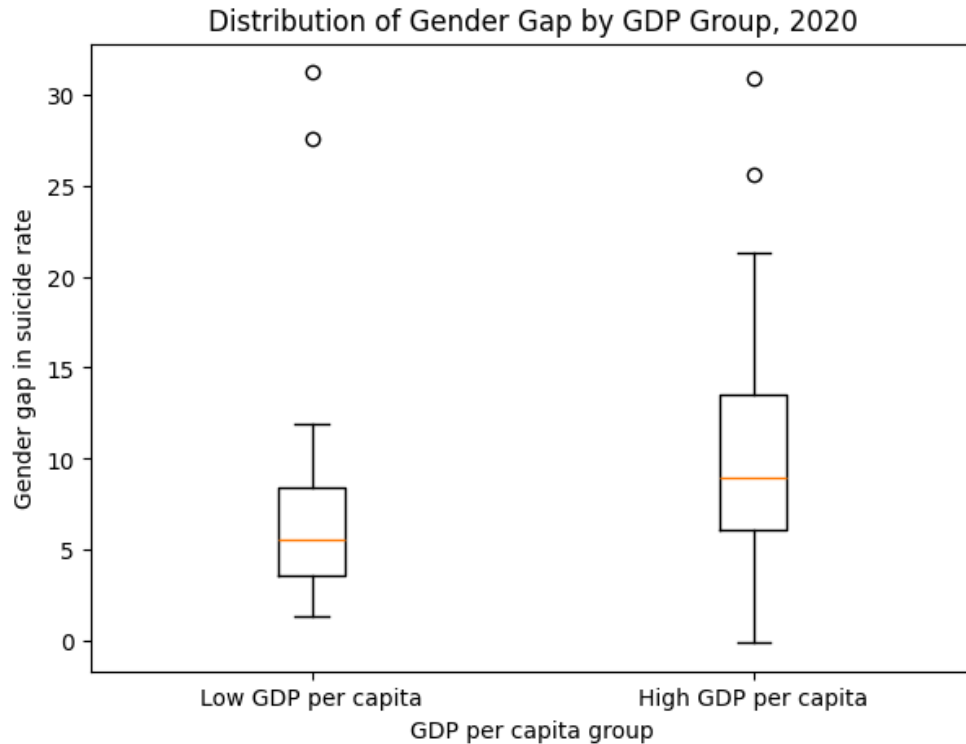


Figure 4. Distribution of gender gap in suicide rates by GDP per capita group, 2020.

Question 3: Which countries deviate most from the expected relationship between GDP per capita and suicide rates in 2020?

Answer:

In the year of 2020 countries that deviate most from the expected relationship are (**Lesotho, Guyana, Suriname, South Korea, Swaziland, Lithuania, Ukraine, Uruguay, South Africa, and Russia**)

Data used:

For this question, we used our combined dataset. This dataset contains the suicide rate and GDP of each country. We mainly used two fields from the dataset “gdp_per_capita_usd” as the x-axis variable and “suicide_rate_both” being the y-axis to find any countries that might deviate from the expected relationship.

Steps and tools used:

For this analysis we used Google Colab including python, pandas, Matplotlib, and scikitlearn.

First, we imported the JSON file using (pd.read_json). Then we start to identify which countries deviate most from the expected results. We created a simple linear regression model first, using the GDP as an independent variable, and suicide rate as the dependent variables. We created this linear model to see if there are any obvious deviate results first, to see whether we can continue with this question. After we determined we could go ahead with the question, we built an 'residual' column, which is (actual suicide rate – expected suicide rate). Counties with the largest residuals were considered that deviated most from the expected relationship.

The key operations executed were:

```
df_2020["residual"] = (  
    df_2020["suicide_rate_both"] -  
    df_2020["expected_suicide_rate"]  
)  
  
df_2020["abs_residual"] = df_2020["residual"].abs()  
  
top_deviations = df_2020.sort_values(  
    "abs_residual",  
    ascending=False  
) .head(10)  
  
top_deviations[  
    [  
        "Country",  
        "gdp_per_capita_usd",  
        "suicide_rate_both",  
        "expected_suicide_rate",  
        "residual",
```

"abs_residual"

]

]

A scatterplot was also created to show the relation, as well double check that the countries we have are correct.

Results :

The top five countries on the lists are (Lesotho, Guyana, Suriname, South Korea, Swaziland) as they have the highest residual. But if we look at the results, we can see that Lesotho indeed has a higher suicide rate than other countries at a similar GDP level, as well South Korea. This again shows that GDP alone cannot fully explain the suicide rate.

Interpretation :

Looking at the result, we can see that most of the biggest deviations were positive residuals. This means that those countries had suicide rates higher than expected if compared to their GDP. This might suggest that economic wealth alone is not enough to explain the suicide rate of a country. We must consider other factors, such as mental health, social pressure, inequality, cultural difference, government structure, war, conflict, and demographic patterns. Therefore, while GDP per capital can be used as one of the explanatory variables, it is not strong enough to predict the results. With this question we can see which countries don't follow the patterns. However, many of them share similar social and historical pressures. For example, some countries have experienced recent war and political instability, like Ukraine, and Russia. Others may face strong social inequality, poverty, unemployment, or limited access to mental health support, such as Lesotho, South Africa, Guyana, and Suriname. And in some countries, cultural expectations, social pressure, problem around mental health, like South Korea. And we know that South Korea also have one of the lowest birth-rates.

Limitations :

The limitations we face in this question are that we only have results from 2020, and that cannot explain long-term trends. Also, we only have results from GDP, and shown above, by itself is not strong enough to build a reliable prediction model.

Country	gdp_per_capita_usd	suicide_rate_both	expected_suicide_rate	residual	abs_residual
Lesotho	1349.98	33.378771	7.897898	25.480873	25.480873
Guyana	5859.75	29.530095	8.155537	21.374558	21.374558
Suriname	6680.51	28.468506	8.202426	20.266080	20.266080
South Korea	34023.56	27.155653	9.764509	17.391145	17.391145
Swaziland	4147.69	23.832791	8.057729	15.775062	15.775062
Lithuania	21159.40	24.516662	9.029591	15.487070	15.487070
Ukraine	3365.13	22.119942	8.013022	14.106920	14.106920
Uruguay	18257.32	22.769361	8.863798	13.905562	13.905562
South Africa	6520.58	21.099576	8.193290	12.906286	12.906286
Russia	11426.09	21.011106	8.473537	12.537570	12.537570

Table 3. Countries with the largest absolute residuals from the expected GDP-suicide relationship in 2020.

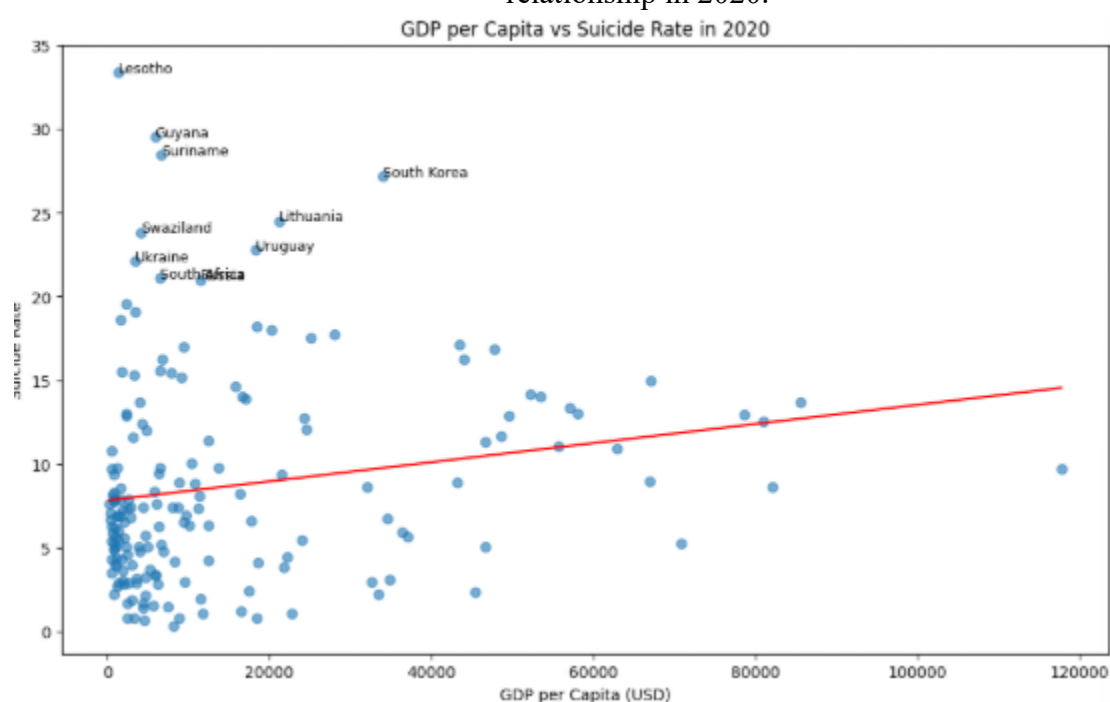


Figure 5. Actual suicide rate and expected GDP-suicide relationship in 2020.